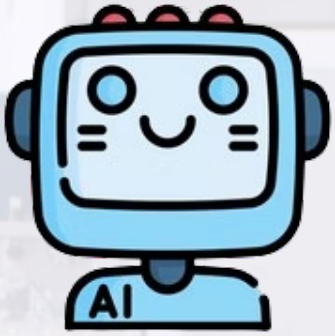
Camp Asimov Student Safety Plan



Robotics Engineering Program

2341 Michigan Ave, Santa Monica, CA
Program Director: Ronit Kumar

1. Our Safety Philosophy

Camp Asimov operates as a supervised engineering lab where students learn to design, build, and program real robotic systems. Safety is a core part of engineering practice and is integrated into every aspect of the program.

Our safety approach is based on three principles:

Supervision

All fabrication activities take place under instructor supervision.

Training

Students receive safety instruction before using any equipment.

Responsibility

Students are expected to treat tools, equipment, and the workspace with respect.

The goal of our safety program is to create an environment where students can explore engineering while maintaining a safe and structured learning environment.

2. Program Environment

Camp Asimov functions similarly to a small engineering lab or makerspace. Students work on projects that may include:

- Robot design and assembly
- CAD (computer-aided design)
- 3D printing of custom parts
- Electronics integration
- Programming and testing robotic systems

> All fabrication tools and equipment are used in accordance with established safety practices.

3. Student Supervision & Safety Orientation

Student Supervision

Students are supervised at all times during camp activities. The program maintains small cohorts to ensure instructors can closely monitor equipment use and provide guidance during fabrication activities.

Instructors are responsible for:

- Demonstrating proper equipment use
- Monitoring tool operation
- Enforcing safety rules
- Stopping any unsafe activity immediately

Students may only use equipment after receiving proper instruction.

Safety Orientation

At the beginning of the program, students participate in a safety orientation covering:

- Workspace expectations
- Tool safety guidelines
- Responsible use of equipment
- Safe behavior around fabrication tools
- Emergency procedures

Students must follow all safety instructions provided by instructors.

4. General Lab Rules & Personal Protective Equipment

General Lab Rules

Students are expected to follow these rules while working in the lab environment:

- Follow instructor directions at all times
- Do not operate equipment without permission
- Keep workspaces clean and organized
- Report damaged equipment or unsafe conditions immediately
- Keep hands clear of moving mechanisms
- Secure loose clothing and long hair when working near equipment

> Unsafe behavior or misuse of tools may result in restricted equipment access.

Personal Protective Equipment

Depending on the activity, students may be required to use protective equipment such as:

- Safety glasses
- Protective gloves when appropriate
- Other protective equipment recommended for specific fabrication processes

Students will receive guidance on when protective equipment is required.

5. Fabrication Equipment

Camp Asimov uses modern fabrication tools commonly found in engineering labs and robotics workshops. All equipment is used in a supervised environment with safety procedures in place.

3D Printers

Students use desktop 3D printers to manufacture custom robot components designed in CAD software. Safety practices include: instructors supervise all printer operation, students avoid contact with heated components, prints are removed only after cooling, and printer enclosures remain closed during operation. Students are taught about the basic operating principles and safety considerations of additive manufacturing systems.

Electronics and Robotics Systems

Students may work with low-voltage robotics electronics such as sensors, motors, control hubs, batteries, and wiring and connectors. Safety procedures include: powering systems off before making modifications, avoiding exposed wiring, using approved batteries and chargers, and instructor supervision during electrical work.

Mechanical Assembly

Students assemble robots using mechanical components including aluminum structural parts, motors and gears, brackets and fasteners, and wheels and drive systems. Students are taught safe practices when assembling mechanical systems and testing moving mechanisms.

6. Future Fabrication Tools & Workspace Organization

Future Fabrication Tools

As the program evolves, Camp Asimov may introduce additional fabrication equipment such as:

- Laser cutters
- CNC machines
- Soldering stations
- Other digital fabrication tools commonly used in engineering labs

Any additional tools will follow the same safety principles: training before use, instructor supervision, and appropriate protective equipment when necessary. Students will only use equipment appropriate for their experience level.

Workspace Organization

Maintaining an organized workspace is essential for safe engineering work. Students are expected to:

- Return tools to their proper location
- Store materials safely
- Keep walkways clear
- Clean work areas at the end of each session

Instructors ensure all equipment is powered down and stored properly at the end of the day.

7. Emergency Procedures & Incident Reporting

Emergency Procedures

In the unlikely event of an emergency:

1. Students must stop work immediately
2. Follow instructor directions
3. Move to designated safe areas if necessary

Camp staff are responsible for responding to emergencies, contacting emergency services when required, and notifying parents or guardians. Emergency contact information is maintained for all students.

Incident Reporting

Any injury, equipment malfunction, or unsafe condition must be reported immediately to an instructor.

Staff will review any incidents to ensure appropriate safety improvements are made.

> Maintaining a safe learning environment is an ongoing process.

8. Parent Acknowledgment

Engineering and robotics activities involve tools, equipment, and fabrication processes. Camp Asimov maintains structured supervision and safety training to minimize risks while allowing students to gain real hands-on engineering experience.

> By participating in the program, students agree to follow all safety guidelines and instructor directions.

Camp Asimov

Robotics Engineering Program
Santa Monica, California

Contact Us

For questions about safety procedures, please contact:
info@campasimov.com